



AUTOPSY

The Complete Digital Forensics Guide

*A Step-by-Step Practical Handbook for Cyber Security Students,
Forensic Investigators & Law Enforcement Professionals*

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Preface

Digital forensics is the backbone of modern cybercrime investigation. When a system is compromised, a crime is committed, or critical evidence must be preserved, digital forensic tools become the investigator's most powerful instruments. Among these tools, Autopsy stands out as the world's most widely used open-source digital forensics platform.

This eBook is written for you — the cyber security student, the aspiring forensic analyst, the law enforcement officer who needs to extract the truth from a digital device. Whether you are encountering Autopsy for the first time or looking to deepen your skills, this guide walks you through every concept, every menu, every module — with real-world context and zero assumptions about prior knowledge.

 **Note:** This guide covers Autopsy 4.x (latest stable). Screenshots and menu paths refer to the Windows desktop version unless otherwise stated.

Chapter 1: Understanding Digital Forensics

1.1 What Is Digital Forensics?

Digital forensics (also called computer forensics or cyber forensics) is the science of identifying, preserving, collecting, analyzing, and presenting digital evidence in a legally sound manner. It answers questions like:

- Who accessed this system, and when?
- What files were created, modified, or deleted?
- Was malware installed on this device?
- What websites did this person visit?
- Did this employee leak confidential data?

Digital forensics is used in criminal investigations, civil litigation, incident response, corporate audits, and intelligence analysis. The evidence gathered must be admissible in court, which demands strict procedures.

1.2 The Digital Forensics Process

Every legitimate forensic investigation follows a structured pipeline:

Phase	Description
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1. Identification	Recognize and scope potential evidence sources — hard drives, mobile devices, cloud backups, network logs.
2. Preservation	Prevent alteration. Create forensic images (bit-for-bit copies). Apply write-blockers. Document chain of custody.
3. Collection	Acquire data using approved methods. Use forensic imaging tools like FTK Imager or dd.
4. Examination	Extract relevant artifacts from images using forensic software (Autopsy, Sleuth Kit, EnCase).
5. Analysis	Correlate findings. Build timelines. Identify malicious activity or evidence of crime.
6. Presentation	Produce court-ready reports. Expert testimony. Export findings with full artifact metadata.

1.3 Types of Digital Evidence

- Disk Images (.E01, .dd, .img) — complete sector-level copies of storage media
- File System Artifacts — MFT records, deleted files, directory structures
- Registry Hives — Windows settings, installed programs, user activity
- Browser History — visited sites, cached pages, download records
- Email Archives — PST/OST files, sent/received messages
- Log Files — event logs, web server logs, firewall logs
- Memory Dumps — volatile RAM captures containing running processes, credentials
- Mobile Device Data — call logs, SMS, app data, GPS coordinates

🚨 **Legal Warning:** Always obtain proper legal authorization before imaging or analyzing any device. Unauthorized forensic examination can constitute a criminal offense in most jurisdictions.

Chapter 2: Introduction to Autopsy

2.1 What Is Autopsy?

Autopsy is a free, open-source graphical interface built on top of The Sleuth Kit (TSK) — a powerful command-line forensics library. Developed and maintained by Basis Technology, Autopsy is trusted by the FBI, US Department of Defense, law enforcement agencies worldwide, and thousands of private investigators.

It provides a complete digital forensics workflow inside a single desktop application: ingest evidence, automatically analyze it with specialized modules, browse artifacts in a familiar UI, build timelines, and generate professional reports.

Feature	Details
License	Open-source (Apache 2.0) — completely free
Platform	Windows (primary), macOS, Linux
Supported Evidence	Disk images, local drives, logical files, memory images, mobile data
Image Formats	E01, Ex01, L01, dd, raw, vmdk, vhd, img, ISO
File Systems	NTFS, FAT12/16/32, exFAT, ext2/3/4, HFS+, APFS, UFS
Plugin System	Java/Python-based ingest modules for extensibility
Report Formats	HTML, Excel, Body File, KML (for geographic data)
Official Website	https://www.autopsy.com

2.2 Autopsy vs Other Forensics Tools

Tool	Comparison with Autopsy
EnCase	Commercial (\$3,500+/year), more advanced enterprise features, but Autopsy is free and nearly as capable for most investigations.
FTK (AccessData)	Commercial, faster indexing engine, but Autopsy's modular pipeline is more flexible.
Volatility	Memory forensics specialist — often used alongside Autopsy, not instead of it.
Wireshark	Network packet analysis — complements Autopsy but doesn't analyze disk images.
CAINE Linux	A live forensics distro that includes Autopsy and many other tools pre-installed.

Chapter 3: Installation & System Setup

3.1 System Requirements

Requirement	Minimum / Recommended
Operating System	Windows 10/11 64-bit (recommended), Ubuntu 18.04+, macOS 10.14+
RAM	Minimum: 4 GB Recommended: 8–16 GB (large images need more)
CPU	Dual-core 2.0 GHz+ Recommended: Quad-core for faster ingest
Storage	At least 2x the size of the evidence image you are analyzing
Java	Autopsy 4.x ships with embedded OpenJDK — no separate install needed
Screen	1280x768 minimum 1920x1080 recommended

3.2 Downloading Autopsy

Follow these steps to download the official Autopsy installer:

1

Open your browser and navigate to the official site

URL: <https://www.autopsy.com/download/>

Always download from the official site to avoid tampered versions.

2

Choose your platform

Windows: Click 'Download for Windows' — this gives you the 64-bit MSI installer.

Linux: Download the .zip package and follow the Linux README.

macOS: Download the .dmg file.

3

Verify the download integrity

On the download page, find the SHA-256 hash for the installer.

After downloading, open PowerShell and run:

```
Get-FileHash autopsy-4.xx.x-64bit.msi -Algorithm SHA256
```

Compare the output hash with the one on the website before installing.

3.3 Installing on Windows (Step-by-Step)

1

Run the Installer as Administrator

Right-click the downloaded .msi file.

Select 'Run as Administrator'.

Click 'Yes' on the UAC prompt.

2

Accept the License Agreement

Read the Apache 2.0 license.

Select 'I accept the terms in the License Agreement'.

Click 'Next'.

3

Choose Installation Directory

Default: C:\Program Files\Autopsy-4.xx.x

Recommendation: Keep the default path unless you have a specific reason to change it.

Click 'Next'.

4

Install

Click 'Install'.
Windows may show a UAC prompt — click 'Yes'.
Wait for the installation to complete (typically 2–5 minutes).

5

Launch Autopsy

Do NOT launch via the 'Finish' checkbox on first install.
Instead, find Autopsy in the Start menu.
Right-click → 'Run as Administrator' — this is important for disk access.

Pro Tip: Always run Autopsy as Administrator, especially when analyzing physical drives or raw images. Without admin rights, Autopsy cannot access raw disk sectors.

3.4 Installing on Ubuntu Linux

Autopsy on Linux requires manual dependency setup. Follow this procedure:

```
# Step 1: Install Java (if not already installed)
sudo apt update
sudo apt install default-jdk -y

# Step 2: Install The Sleuth Kit dependencies
sudo apt install sleuthkit -y

# Step 3: Download Autopsy Linux ZIP from autopsy.com
# Extract and move to /opt
unzip autopsy-4.xx.x.zip
sudo mv autopsy-4.xx.x /opt/autopsy

# Step 4: Run the setup script
cd /opt/autopsy
chmod +x unix_setup.sh
sudo ./unix_setup.sh

# Step 5: Launch Autopsy
cd bin/
sudo ./autopsy
```

Chapter 4: Autopsy User Interface — A Complete Tour

4.1 Welcome Screen

When you first launch Autopsy, you see three main options on the Welcome Screen:

Option	When to Use It
New Case	Starting a brand new investigation from scratch.
Open Case	Resuming work on a previously created investigation (.aut file).
Open Recent Case	Quick access to your last 5 recently opened cases.

4.2 Main Application Layout

Once a case is open and evidence is loaded, the Autopsy interface is divided into five main regions:

Region	Description
Tree View (Left Panel)	Hierarchical view of your case — data sources, file system nodes, artifact categories, and ingest results.
Result Viewer (Top Right)	Displays the contents of whatever is selected in the Tree View — files, records, search results.
Content Viewer (Bottom Right)	Shows the actual content of the selected item — hex, strings, images, metadata, text, application views.
Keyword Search Bar (Top)	Run keyword searches across all indexed content in the case.
Status Bar (Bottom)	Shows ingest progress, active background jobs, and system status messages.

4.3 Tree View — Navigating Your Evidence

The Tree View is your primary navigation tool. It is organized as follows:

- Case Name (root node)
 - Data Sources — each added image or drive appears here
 - Volume / Partition
 - File System → Root Folders → Files
- Views — organized cross-cutting views of artifacts
 - File Types (Images, Videos, Documents, Executables...)
 - Deleted Files
 - By Extension / By MIME Type
- Results — output from ingest modules
 - Extracted Content (Web History, Recent Docs, Installed Programs...)

- Keyword Hits
- Hashset Hits
- Email Messages
- Tags — any file or artifact you have manually tagged
- Reports — generated reports appear here when complete

Chapter 5: Creating Your First Case

5.1 Case Concepts

In Autopsy, a Case is the top-level container for an entire investigation. A case can contain multiple data sources (disk images, drives, logical files). All analysis results, tags, and reports are stored inside the case. Case data is saved in a folder you specify, plus an .aut project file.

Best Practice: Create one case per investigation. Use a consistent naming convention: YYYYMMDD_CaseID_SubjectLastName (e.g., 20250601_INV042_Sharma). Store cases on a dedicated forensic workstation, not on the same drive as the evidence.

5.2 Step-by-Step: Creating a New Case

1**Open New Case Wizard**

From the Welcome Screen, click 'New Case'.
Alternatively: File menu → New Case.

2**Enter Case Information**

Case Name: A descriptive name (e.g., USB_Investigation_2025).
Base Directory: The folder where Autopsy will store all case files.

- Use a dedicated forensics partition, not C:\
- Ensure you have at least 2x the evidence size in free space.

Click 'Next'.

3**Enter Optional Case Details**

Case Number: Your official investigation reference number.
Examiner Name: Your full name (appears in reports).
Examiner Phone / Email: Contact information for the report.
Organization Name: Your agency or institution (e.g., CT University CyberForensics Lab).
Case Notes: Brief description of the investigation purpose.

Click 'Finish'.

4

Add Data Source
Autopsy immediately opens the 'Add Data Source' dialog.
This is covered in full detail in Chapter 6.
You can also skip and add evidence later via File → Add Data Source.

5.3 Case File Structure

Autopsy creates a folder structure like this on disk:

```
USB_Investigation_2025/
├── USB_Investigation_2025.aut      ← Main project file
├── ModuleOutput/                 ← Results from ingest modules
│   ├── keyword_search/
│   ├── file_type/
│   └── hash_lookup/
├── Reports/                      ← Generated HTML/Excel reports
└── Cache/                       ← Thumbnail cache, text index
```

Chapter 6: Adding Evidence (Data Sources)

6.1 What Is a Data Source?

A data source is any piece of digital evidence you add to your Autopsy case for analysis. Autopsy supports four types of data sources:

Data Source Type	Description & Use Case
Disk Image / VM File	Forensic images (.E01, .dd, .img, .vmdk, .vhd). The most common input — bit-for-bit copies of physical drives.
Local Disk	Analyze a live physical drive connected to your forensics workstation. Best used with a hardware write-blocker attached.
Logical Files	A folder or individual files — useful when you cannot image the whole drive (e.g., analyzing a USB's file dump, a folder export).
Autopsy Logical Imager Result	Results from Autopsy's field triage tool — a lightweight collector that grabs targeted files from a live system.

6.2 Adding a Disk Image (Most Common Workflow)

1

Open Add Data Source Dialog

File menu → Add Data Source

— or — click the 'Add Data Source' button in the toolbar.

The wizard opens. Select 'Disk Image or VM File' and click 'Next'.

2

Browse for the Image File

Click 'Browse' and navigate to your .E01 or .dd image.

If the image is split into segments (e.g., image.E01, image.E02...),

select only the FIRST segment — Autopsy auto-detects the rest.

Time Zone: Select the timezone of the original system if known.

This ensures timestamps are displayed correctly.

Click 'Next'.

3

Configure Ingest Modules

This is a critical step — covered in full detail in Chapter 7.

For now, select the modules you want to run automatically.

Recommendation for first-time analysis: select all modules.

Click 'Next'.

4

Review and Add

Autopsy shows a summary of your selected image and modules.

Click 'Finish'.

Autopsy begins ingesting the image — this may take minutes to hours

depending on image size and selected modules.

You can browse the file system immediately while ingest continues in background.

6.3 Adding a Local Drive (Live Disk Analysis)

Analyzing a live drive is riskier than analyzing an image because there is a chance of modifying evidence. Always use a hardware write-blocker when connecting a suspect's drive.

1

Connect the drive

Attach the suspect's drive via a write-blocker (e.g., Tableau T8u, WiebeTech).

The write-blocker prevents any writes to the suspect drive.

2

Add Data Source → Local Disk

File → Add Data Source → Local Disk.

Autopsy lists all connected drives (identified by disk number).

Select the correct drive carefully — wrong drive = wrong evidence.

Check 'Create an image copy of the disk' to make a forensic backup first (recommended).

3

Continue as with disk image

Select ingest modules and proceed as in Section 6.2 Steps 3–4.

⚠ Critical: NEVER connect a suspect drive to a forensics workstation without a hardware write-blocker. Software write-blocking is unreliable. Evidence contamination can invalidate your entire investigation.

Chapter 7: Ingest Modules — The Brain of Autopsy

7.1 What Are Ingest Modules?

Ingest modules are automated analysis plugins that run against your evidence when you add a data source. Each module specializes in one type of analysis — hash lookup, keyword search, browser history extraction, and so on. They run in parallel in the background and populate the Results tree with findings.

There are two types:

- File-level ingest modules — process each file individually (e.g., Hash Lookup, EXIF Parser)
- Data-source-level ingest modules — process the entire data source (e.g., PhotoRec Carver, Email Parser)

7.2 Core Ingest Modules Reference

Module	What It Does
Recent Activity	Extracts web browsing history (Chrome, Firefox, IE/Edge), recently opened documents, installed programs, device attachments, shutdown times, and user accounts from the Windows Registry.
Hash Lookup	Computes MD5 and SHA-256 hashes of every file, then checks them against NSRL (National Software Reference Library) and custom hash sets to flag known-bad (malware) or known-good (OS) files.
File Type Identification	Identifies files by their true MIME type (not just extension) using the Apache Tika library. Catches renamed malware (e.g., virus.jpg that is actually an .exe).

Embedded File Extractor	Unpacks archive files (ZIP, RAR, 7z), Office documents with embedded content, and email attachments — then processes the extracted files through all other modules.
EXIF Parser	Extracts metadata from image files (JPEG, TIFF) — GPS coordinates, camera model, timestamp, software used. Critical for geolocating photos.
Keyword Search	Performs a full-text index of all files and memory strings, enabling keyword and regular-expression searches. Can use pre-defined lists (credit card numbers, phone numbers, IP addresses, URLs, email addresses).
Email Parser	Extracts and indexes email messages from PST (Outlook), Thunderbird (.msf), Mbox, and EML files. Displays sender, recipient, date, subject, and body.
Extension Mismatch Detector	Flags files where the extension does not match the true file type — a common technique used to hide malicious files.
E01 Verifier	Verifies the MD5 hash of E01 image segments to confirm image integrity has not been compromised.
Android Analyzer	Parses Android SQLite databases — call logs, SMS, MMS, contacts, browser history, GPS data.
PhotoRec Carver	Carves deleted files from unallocated space using file signature patterns — can recover JPEGs, documents, and more even after deletion.
Plaso / Log2Timeline	Timeline generation — creates a super-timeline from all timestamp artifacts in the image.
Interesting Files Identifier	Matches files against configurable rule sets to flag suspicious or relevant files automatically.
Virtual Machine Extractor	Detects and extracts virtual machine disk images (VMDK, VHD) found within the evidence and adds them as nested data sources.
Data Source Integrity	Verifies the hash of the entire data source on ingest to ensure evidence integrity.

7.3 Configuring the Hash Lookup Module

The Hash Lookup module is one of the most important for triage. Setting it up correctly can immediately flag known malware.

1

Open Hash Database Management

Tools menu → Options → Hash Sets.

Or during ingest module configuration, click 'Hash Lookup' → 'Configure'.

2

Import NSRL (Known-Good Hashes)

Download the NSRL hash set from: <https://www.nist.gov/itl/ssd/software-quality-group/national-software-reference-library-nsrl>

Click 'Import Database'.

Select the downloaded NSRLFile.txt.
Set Type to 'Known Good'. Click 'OK'.

3

Add Custom Hash Sets (Known Bad)

If you have IOC hash lists (from VirusTotal, CISA, FBI, etc.):
Click 'New Hash Set' → name it (e.g., 'Known_Malware_2025').
Set Type to 'Notable (Known Bad)'.
Import your hashes. Click 'OK'.
Files matching this set will be flagged in red in results.

7.4 Configuring Keyword Search

1

Access Keyword Lists

During ingest setup, click on 'Keyword Search' → 'Configure'.
— or — Tools → Options → Keyword Search.

2

Enable Built-in Lists

Autopsy includes ready-made lists:

- Credit Card Numbers (regex)
- Phone Numbers (regex)
- IP Addresses (regex)
- Email Addresses (regex)
- URLs

Check the boxes next to lists relevant to your investigation.

3

Create Custom Keyword Lists

Click 'New List'.

Name it (e.g., 'Suspect_Keywords').

Add keywords — exact match or regex.

Examples:

- Exact: 'confidential', 'password', suspect's name
- Regex: `\b\d{4}[- ]?\d{4}[- ]?\d{4}[- ]?\d{4}\b` (credit card pattern)

Click 'Save'.

Chapter 8: Browsing & Analyzing Evidence

8.1 The File System Browser

Once evidence is ingested, you can browse the complete file system tree just like Windows Explorer. In the Tree View, expand your data source → partition → file system → folders.

Key things to notice:

- Deleted files appear with a red X icon — they were found in unallocated space or MFT records.
- Files with a flag icon have been flagged by an ingest module (e.g., hash match, keyword hit).
- Right-clicking any file shows options: Tag, Extract, Add to Report, View in Timeline, Hash Lookup.

8.2 The Content Viewer — Examining Files

When you click on a file in the Result Viewer, the Content Viewer at the bottom shows different views via tabs:

Tab	Description
Hex	Raw hexadecimal dump of the file — essential for identifying file signatures, obfuscated content, and embedded objects.
Strings	Extracts all printable ASCII and Unicode strings from the file — often reveals URLs, registry keys, passwords, and malicious commands inside binaries.
File Metadata	Shows inode/MFT metadata — file size, timestamps (Created, Modified, Accessed, MFT Entry Changed), allocated/unallocated status, parent folder.
Results	Lists all ingest module results for this file — keyword hits, hash hits, EXIF data, etc.
Text	Rendered text content — useful for documents and logs.
Application	Rendered view — images display as images, HTML renders in a browser pane.
Indexed Text	Full indexed text extracted from the file for keyword searching.
Other Occurrences	Shows if the same hash appears in other cases or data sources in your system.

8.3 The Results Tree — Working with Artifacts

The Results section of the Tree View is where all automatically extracted artifacts appear. Here is what each category contains:

Results Category	Artifacts Inside
Extracted Content → Web History	URLs visited, visit dates, browser used, page titles. Chrome, Firefox, IE, Edge all supported.
Extracted Content → Web Downloads	Files downloaded from the internet — filename, source URL, date, local save path.

Extracted Content → Web Cookies	Cookie name, value, domain, expiry, creation date.
Extracted Content → Web Bookmarks	Bookmarked pages and their URLs.
Extracted Content → Web Cache	Cached web page fragments stored by the browser.
Extracted Content → Web Search	Search queries typed into Google, Bing, and other search engines.
Extracted Content → Recent Documents	Files recently opened by the user (from Windows Registry MRU lists).
Extracted Content → Installed Programs	Software installed on the system — name, publisher, install date (from Registry).
Extracted Content → OS Accounts	Windows user accounts, SIDs, login names.
Extracted Content → OS Info	Operating system name, version, install date, last login, timezone.
Extracted Content → USB Device Attached	USB devices ever connected — vendor, product, serial number, first/last connection time.
Extracted Content → Run Programs	Programs listed in Registry Run keys — startup persistence mechanisms.
Extracted Content → EXIF Metadata	GPS coordinates, camera model, timestamps from photos.
Extracted Content → Email Messages	Full email messages parsed from PST/Mbox/EML files.
Keyword Hits	Files containing your keyword search matches — sorted by list name.
Hashset Hits	Files whose hashes matched known-good or known-bad hash sets.
Interesting Items	Files matching the Interesting Files Identifier rules.

Chapter 9: Advanced Features

9.1 Timeline Analysis

The Timeline is one of Autopsy's most powerful features. It builds a chronological view of every timestamped event in your evidence — file creation, modification, access, registry writes, web visits, email arrivals, and more.

1

Open the Timeline

Tools menu → Timeline.

Autopsy opens the Timeline window (may take a moment to load all events).

2

Choose View Mode

Counts (Bar Chart): Shows event density over time — great for spotting spikes in activity.

Details: Shows individual events in a gantt-style view — drill into specific hours.

List: Flat chronological list of all events — useful for export and examination.

3

Apply Filters

Left panel allows filtering by:

- Event Type (File System, Registry, Web Activity, EXIF, etc.)
- Data Source (if you have multiple images)
- Tags (only show tagged items)
- Text Search (filter events containing specific text)

4

Identify Suspicious Activity

Look for unusual spikes — e.g., mass file deletions at a specific time.

Look for activity outside normal working hours.

Look for file timestamps inconsistent with other evidence (possible anti-forensics).

Right-click any event → 'View File in Directory' to jump to it in the file browser.

Right-click → 'Add Tag' to bookmark it for your report.

9.2 Keyword Search

Even after ingest, you can run ad-hoc keyword searches at any time.

1

Open Keyword Search

Click the magnifying glass icon in the top toolbar.
— or — Tools → Keyword Search.

2

Enter Your Query

Type your keyword in the search box.

Toggle 'Substring' to search for partial matches.

Toggle 'Regex' to use regular expression patterns.

Example regex for Indian mobile numbers: `[6-9]\d{9}`

3

Review Results

Results appear in the Result Viewer showing filename, path, hit count, and snippet.

Click any result to view the file in Context Viewer.

Use 'Previous/Next' arrows to navigate between hits within a single file.

9.3 File Carving with PhotoRec

File carving recovers deleted files from unallocated disk space by looking for known file signatures (magic bytes), even when the file system entries are gone.

1**Enable the PhotoRec Carver Module**

When adding a data source, check 'PhotoRec Carver' in ingest modules.
— or — Run it post-ingest: right-click data source → Run Ingest Modules → select PhotoRec Carver.

2**Review Carved Files**

Results appear under: Views → File Types → By Type (all recovered file types).
Carved files are listed with '(carved)' indicator.
Note: carved files may not have original filenames — only the type and content.

3**Filter and Analyze**

Focus on carved images first — they can contain explicit content, screenshots, or evidence photos.
Carved documents may contain text, spreadsheets, or database fragments.
Tag relevant findings for the report.

9.4 Tagging Evidence

Tagging marks files or artifacts as significant for your investigation. Tags appear in reports and can be searched.

- Right-click any file → Tag File → choose an existing tag or create a new one.
- Built-in tags include: Notable Item, Follow Up, Bookmark, Bad (Evidence).
- Create custom tags: Tools → Options → Tags → New Tag.
- Tagged items appear under the 'Tags' node in the Tree View.
- All tags are included in the HTML and Excel reports with full metadata.

📌 Best Practice: Tag evidence as you find it — don't wait until the end. Include a note with each tag explaining its investigative significance. This saves time when writing the final report.

9.5 Communications Analysis

Autopsy's Communications feature (available under Tools → Communications) shows a social graph of all contacts, calls, SMS, email, and chat extracted from mobile devices or mail archives.

- Filter by date range to isolate communications during the incident window.
- Click on a contact node to see all messages to/from them.
- Export the graph as a visual for court presentations.

Chapter 10: Deleted File Recovery

10.1 How Files Are Deleted

Understanding deletion is fundamental to forensic recovery. When a file is 'deleted' in Windows:

- The file's MFT (Master File Table) entry is marked as unallocated — not erased.
- The disk sectors occupied by the file are marked as free — but not overwritten.
- The actual data remains on disk until those sectors are reused by new data.

Even files deleted from the Recycle Bin, or with Shift+Delete, follow this pattern. Only secure deletion tools (which overwrite sectors with zeros or random data) truly destroy files.

10.2 Viewing Deleted Files in Autopsy

1**Navigate to Deleted Files**

In the Tree View, go to: Data Source → File System → '\$OrphanFiles'
— or — Views → Deleted Files.
Files with a red X badge are marked as deleted in the file system.

2**Examine Deleted File Content**

Click on a deleted file.
If the data is still intact (sectors not overwritten), the Content Viewer will show the content.
If only partial data remains, the hex view will show zeroed or garbage sections.
Check the 'File Metadata' tab for original path and timestamps.

3**Extract Deleted Files**

Right-click the file → Extract File(s).
Choose destination folder.
Autopsy copies the recovered data to that folder.
For multiple files: select them all (Ctrl+Click) → right-click → Extract File(s).

10.3 Recovering Files from Unallocated Space

Files completely removed from the MFT can still be recovered from unallocated space using the PhotoRec Carver module. See Section 9.3 for the full procedure.

🔍 Forensic Insight: The more a device has been used after deletion, the lower the recovery success rate. A suspect who used the drive heavily after deleting evidence is harder to recover from. Act quickly when preserving evidence — every write to the suspect device reduces your chances.

Chapter 11: Generating Reports

11.1 Why Reports Matter

A forensic report is the formal, legal document that presents your findings to lawyers, judges, clients, or management. It must be accurate, objective, reproducible, and clearly written. Autopsy's reporting engine automates the data collection — you provide the professional framing.

11.2 Generating an HTML Report

1**Open Report Wizard**

Go to: Generate Report button in the toolbar
— or — Tools → Generate Report.

2**Select Report Module**

HTML Report: Best for viewing in a browser; includes thumbnails, links, tables.
Excel/CSV: Best for data analysis and import into other tools.
Body File: For timeline correlation with external tools (Plaso, log2timeline).
KML: For geographic data visualization in Google Earth.
Select 'HTML Report' for a comprehensive professional report.

3**Configure Report Scope**

All Results: Full report of all artifacts.
Tagged Results Only: Only items you tagged (recommended for court reports).
Select which artifact types to include (Web History, Files, Hash Hits, Tags, etc.).
Click 'Next'.

4**Set Report Name and Location**

Autopsy defaults to saving in the case's Reports/ folder.

Enter a descriptive name: e.g., 'Final_Investigation_Report_2025_06_01'.
Click 'Finish'.

5**Open and Review the Report**

When generation completes, Autopsy offers to open the report.

The HTML report opens in your browser.

It includes: case metadata, examiner name, data source details, and all artifacts in organized tables.

Verify all key findings appear correctly before submitting.

11.3 Professional Report Writing Tips

- Always include your chain of custody documentation as an attachment.
- State your methodology — which modules ran, which hash sets were used, what keywords were searched.
- Distinguish between what the evidence PROVES and what it merely SUGGESTS.
- Use neutral, objective language. Avoid words like 'obviously' or 'clearly'.
- Document negative findings too — files searched but not found, modules that found nothing.
- Include the hash of the evidence image to prove it was not modified during analysis.

Chapter 12: Real-World Investigation Scenarios

Scenario 1: USB Drive Investigation

Situation: An employee is suspected of copying confidential company files to a personal USB drive before resigning.

1. Acquire a forensic image of the USB drive using FTK Imager or Autopsy's built-in imager.
2. Create a new case: USB_Exfiltration_2025.
3. Add the image as a data source. Enable: Recent Activity, Hash Lookup, Keyword Search, File Type Identification.
4. In Tree View: Extracted Content → USB Device Attached — confirm device was connected to the employee's workstation.
5. Browse the file system for files with company-related names or extensions (.docx, .xlsx, .pdf, .pptx).
6. Check deleted files — suspect may have tried to erase evidence of copying.
7. Run keyword search for company name, project names, client names, 'confidential'.
8. Check EXIF data on any images — may contain metadata linking them to company devices.
9. Tag all relevant files. Generate a Tagged Results HTML report.

Scenario 2: Malware Investigation

Situation: A corporate workstation is suspected of being infected with malware. The drive has been imaged and handed to you.

- 10. Add the image. Enable ALL ingest modules, especially Hash Lookup with a known-malware hash set.
- 11. Check Hashset Hits immediately — any known-bad hashes are flagged in red.
- 12. Go to Extracted Content → Run Programs — check all Registry Run keys for suspicious executables.
- 13. Extracted Content → Installed Programs — look for unknown or recently installed software.
- 14. Run keyword search for common malware strings: 'cmd.exe /c', 'powershell -enc', 'certutil', 'regsvr32'.
- 15. Check Extension Mismatch results — malware often masquerades as image or document files.
- 16. Timeline analysis: find the moment of infection — a spike in file creation at an unusual time.
- 17. Extracted Content → Web History — identify the infection vector (phishing link, malicious download).

Scenario 3: Mobile Device Data Recovery

Situation: A victim's Android phone needs to be analyzed for evidence of harassment or cyberstalking.

- 18. Acquire a logical or physical dump of the Android device using UFED, Cellebrite, or ADB backup.
- 19. Add the dump as a Logical Files data source in Autopsy.
- 20. Enable Android Analyzer and Keyword Search modules.
- 21. Extracted Content → Contacts — build a contact profile.
- 22. Extracted Content → Call Log — look for repeated calls from specific numbers.
- 23. Extracted Content → SMS/MMS Messages — export threatening messages as tagged items.
- 24. EXIF Parser results — photos may contain GPS coordinates proving the suspect's location.
- 25. Web History — check if the suspect visited the victim's social media profiles obsessively.

Chapter 13: Best Practices & Legal Compliance

13.1 The Golden Rules of Digital Forensics

Rule	Why It Matters
Never work on original evidence	Always analyze a verified forensic copy. The original is preserved for court.
Document everything	Time-stamp your actions. Courts will question any undocumented procedure.

Verify image integrity	Always compare MD5/SHA-256 hash of your working copy to the original before analysis.
Use write-blockers	Hardware write-blockers for physical media. Never plug a suspect drive without one.
Maintain chain of custody	Record who had possession of evidence, when, where, and why.
Remain objective	You are a finder of fact, not a prosecutor. Report what the evidence shows — including exculpatory findings.
Secure your case files	Encrypt case data at rest. Use ACLs to restrict access. Evidence leaks can cause mistrials.

13.2 Relevant Indian Laws

Forensic practitioners in India must be aware of these key legal provisions:

Act / Section	Relevance
IT Act 2000, Section 65B	Admissibility of electronic evidence in court — requires a certificate from the person responsible for the computer/device.
IT Act 2000, Section 43	Civil penalties for unauthorized access to computer systems.
IT Act 2000, Section 66	Criminal penalties for computer-related fraud and offenses.
IPC Section 420	Cheating and dishonesty — applicable to cybercrime cases.
IPC Section 463/464	Forgery — digital document tampering.
CERT-In Guidelines 2022	Mandatory incident reporting timelines and security standards for Indian organizations.

 **For ForensicAid:** The modules you are building in ForensicAid align precisely with these legal requirements — especially the report generation with Section 65B certification fields, which can make the difference between admissible and inadmissible digital evidence in Indian courts.

Chapter 14: Troubleshooting Common Issues

Problem	Solution
Autopsy crashes on launch	Run as Administrator. Ensure Java is not conflicting — Autopsy uses its own embedded JRE. Check Windows Event Viewer for crash logs.
Ingest modules stuck at 0%	Large images take time. Check Task Manager — if disk I/O is active, it's working. If no disk activity for >10 min, cancel and retry.
'Cannot read image' error	Ensure the full path to the image has no spaces or special characters. Try moving the image to C:\ForensicImages\.

Timeline shows incorrect timestamps	Set the correct timezone for the data source when adding it. You can change it later: right-click data source → Properties.
Keyword search returns no results	Ensure the Keyword Search module ran during ingest. If not, run it post-ingest: right-click data source → Run Ingest Modules → Keyword Search.
Deleted files show no content	Sectors were overwritten by new data after deletion. Use PhotoRec carving for a last attempt. Otherwise, data is unrecoverable.
Report generation hangs	Reduce scope — try generating a Tagged Results Only report. Large cases with millions of artifacts can take hours to report.
Out of disk space during ingest	Autopsy's text index can be large. Ensure your case drive has 3–5x the image size available.

Chapter 15: Keyboard Shortcuts & Quick Reference

Shortcut / Action	Function
Ctrl+F	Open Keyword Search
Ctrl+T	Open Timeline
Ctrl+Shift+N	New Case
Ctrl+O	Open Case
Ctrl+R	Generate Report
F3	Find next keyword hit
Right-click file → Tag	Tag current file
Right-click file → Extract	Extract file to disk
Right-click file → View in Timeline	Jump to file's timestamp in Timeline
Ctrl+A (Result Viewer)	Select all items
Right-click → Export CSV	Export result table to CSV
Alt+Left / Alt+Right	Navigate back/forward in selection history

Appendix: Useful Resources

Official Documentation & Downloads

- Autopsy Official: <https://www.autopsy.com>
- Autopsy User Docs: <https://sleuthkit.org/autopsy/docs/user-docs/4.19.3/>
- The Sleuth Kit (Autopsy's core): <https://www.sleuthkit.org>

- NIST NSRL Hash Sets: <https://www.nist.gov/itl/ssd/software-quality-group/national-software-reference-library-nsrl>
- Basis Technology Training: <https://www.basistech.com/digital-forensics/>

Practice Resources & CTF Labs

- Digital Forensics CTF — CyberDefenders: <https://cyberdefenders.org>
- DFIR.training challenge images: <https://dfir.training>
- BlueTeamLabs Online: <https://blueteamlabs.online>
- Autopsy sample disk images: <https://digitalcorpora.org>

Complementary Tools

- FTK Imager (free imaging tool): <https://www.exterro.com/ftk-imager>
- Volatility 3 (memory forensics): <https://github.com/volatilityfoundation/volatility3>
- Wireshark (network analysis): <https://www.wireshark.org>
- CAINE Linux (forensics live distro): <https://www.caine-live.net>
- RegRipper (deep Windows Registry analysis): <https://github.com/keydet89/RegRipper3.0>

Indian Cyber Law & CERT-In References

- CERT-In: <https://www.cert-in.org.in>
- IT Act 2000 (full text): <https://www.meity.gov.in/content/information-technology-act>
- Cyber Crime Portal India: <https://cybercrime.gov.in>

About the Author

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This eBook is dedicated to every investigator who believes that truth, preserved correctly, always finds its way to justice.

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